

The following are my overall representative publications that I am truly proud of. The first two represent my recent work as a faculty member at UofSC, while the last two represent the work I have done during my postdoctoral training at Imperial College London and CMU. I also provided the links to the open-access versions of the papers.

1. (2023) **FlexiBO**: A cost-aware multi-objective Bayesian optimization algorithm. ¹
2. (2022) **Unicorn**: A causal reasoning approach for performance debugging of configurable systems. ²
3. (2019) **MARS**: an integrated learning and quantitative planning approach that enables runtime reconfiguration of autonomous robots. ³
4. (2016) **BO4CO**: A method for multi-objective optimization of stream processing systems. ⁴

1: Md Shahriar Iqbal, Jianhai Su, Lars Kotthoff, and **Pooyan Jamshidi**. 'FlexiBO: A Decoupled Cost-Aware Multi-Objective Optimization Approach for Deep Neural Networks'. In: Journal of Artificial Intelligence Research (**JAIR**). 2023

2: Md Shahriar Iqbal, Rahul Krishna, Mohammad Ali Javidian, Baishakhi Ray, and **Pooyan Jamshidi**. 'Unicorn: Reasoning about Configurable System Performance through the Lens of Causality'. In: European Conference on Computer Systems (**EuroSys**). 2022

3: **Pooyan Jamshidi**, Javier Cámara, Bradley Schmerl, Christian Kaestner, and David Garlan. 'Machine Learning Meets Quantitative Planning: Enabling Self-Adaptation in Autonomous Robots'. In: 14th IEEE/ACM International Symposium on Software Engineering for Adaptive and Self-Managing Systems (**SEAMS**). 2019

4: **Pooyan Jamshidi** and Giuliano Casale. 'An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems'. In: Modeling, Analysis and Sim. of Computer and Telecomm. Systems (**MASCOTS**). 2016

Here is a list of representative publications based on the communities that I am involved in.

Computer Systems:

1. Md Shahriar Iqbal, Ziyuan Zhong, Iftikhar Ahmad, Baishakhi Ray, and **Pooyan Jamshidi**. 'CAMEO: A Causal Transfer Learning Approach for Performance Optimization of Configurable Computer Systems'. In: ACM Symposium on Cloud Computing (**SoCC**). 2023
2. Md Shahriar Iqbal, Rahul Krishna, Mohammad Ali Javidian, Baishakhi Ray, and **Pooyan Jamshidi**. 'Unicorn: Reasoning about Configurable System Performance through the Lens of Causality'. In: European Conference on Computer Systems (**EuroSys**). 2022
3. Saeid Ghafouri, Kamran Razavi, Mehran Salmani, Alireza Sanaee, Tania Lorido-Botran, Lin Wang, Joseph Doyle, and **Pooyan Jamshidi**. 'IPA: Inference Pipeline Adaptation to Achieve High Accuracy and Cost-Efficiency'. In: Journal of Systems Research (**JSys**). 2023
4. **Pooyan Jamshidi** and Giuliano Casale. 'An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems'. In: Modeling, Analysis and Sim. of Computer and Telecomm. Systems (**MASCOTS**). 2016

Software Engineering:

1. **Pooyan Jamshidi**, Norbert Siegmund, Miguel Velez, Christian Kaestner, Akshay Patel, and Yuvraj Agarwal. 'Transfer Learning for Performance Modeling of Configurable Systems: An Exploratory Analysis'. In: Int'l Conf. Automated Software Engineering (**ASE**). 2017
2. Miguel Velez, **Pooyan Jamshidi**, Norbert Siegmund, Sven Apel, and Christian Kaestner. 'White-box analysis over machine learning: Modeling performance of configurable systems'. In: Proc. Int'l Conf. on Software Engineering (**ICSE**). 2021

Robotics:

1. Md Abir Hossen, Sonam Kharade, Bradley Schmerl, Javier Cámara, Jason M O'Kane, Ellen C Czaplinski, Katherine A Dzurilla, David Garlan, and **Pooyan Jamshidi**. 'CaRE: Finding Root Causes of Configuration Issues in Highly-Configurable Robots'. In: IEEE Robotics and Automation Letters (**RA-L** and **IROS**). 2023
2. Md Abir Hossen, Sonam Kharade, Bradley Schmerl, Jason M O'Kane, David Garlan, and **Pooyan Jamshidi**. 'CURE: Simulation-Augmented Auto-Tuning in Robotics'. In: IEEE Transactions on Robotics (**T-RO**). 2023
3. **Pooyan Jamshidi**, Javier Cámara, Bradley Schmerl, Christian Kaestner, and David Garlan. 'Machine Learning Meets Quantitative Planning: Enabling Self-Adaptation in Autonomous Robots'. In: 14th IEEE/ACM International Symposium on Software Engineering for Adaptive and Self-Managing Systems (**SEAMS**). 2019
4. **Pooyan Jamshidi**, Miguel Velez, Christian Kaestner, Norbert Siegmund, and Prasad Kawthekar. 'Transfer Learning for Improving Model Predictions in Highly Configurable Software'. In: Software Engineering for Adaptive and Self-Managing Systems (**SEAMS**). 2017

AI/ML:

1. Md Shahriar Iqbal, Jianhai Su, Lars Kotthoff, and **Pooyan Jamshidi**. 'FlexiBO: A Decoupled Cost-Aware Multi-Objective Optimization Approach for Deep Neural Networks'. In: Journal of Artificial Intelligence Research (**JAIR**). 2023
2. Mohammad Ali Javidian, Marco Valtorta, and **Pooyan Jamshidi**. 'Learning LWF Chain Graphs: A Markov Blanket Discovery Approach'. In: Conference on Uncertainty in Artificial Intelligence (**UAI**). PMLR. 2020
3. Mohammad Ali Javidian, Marco Valtorta, and **Pooyan Jamshidi**. 'AMP Chain Graphs: Minimal Separators and Structure Learning Algorithms'. In: Journal of Artificial Intelligence Research (**JAIR**). 2020